

Pradyumna Sadhu

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EDUCATION

• University of California, Riverside

Masters and Ph.D. in Astronomy

Sept 2022 - Present

Riverside, CA, USA

- Advisor: Prof. Laura Sales
- CGPA: 4.0/4.0
- Dean's Distinguished Fellowship recipient

• Indian Institute of Technology, Hyderabad

Bachelor of Technology with Honors in Engineering Physics

Aug 2018 - May 2022

Hyderabad, India

- CGPA: 9.5/10.0
- Silver Medal for highest CGPA in the graduating class
- Research Advisors: Prof. Shantanu Desai (IIT Hyderabad, India), Dr. Yong Tian (NCU, Taiwan)

PUBLICATIONS

Three Key Publications:

1. **Pradyumna Sadhu**, Laura V. Sales, Julio F. Navarro, Raphaël Errani, Jose Benavides, Eric W. Peng.
The abundance and radial distribution of faint and ultra-faint dwarfs in galaxy clusters.
MNRAS, Accepted. DOI: 10.1093/mnras/stag1604.
2. **Pradyumna Sadhu**, Yong Tian (2024).
The Baryonic Faber Jackson Relation in Galaxy Groups.
MNRAS, Vol. 528, Issue 4, pp. 5612–5623. DOI: 10.1093/mnras/stae343
3. **Pradyumna Sadhu**, Sajal Gupta, Sowmya Seeram, Shantanu Desai (2021).
Yet another test of Radial Acceleration Relation for galaxy clusters.
Physics of the Dark Universe, Vol. 31. DOI: 10.1016/j.dark.2020.100765

Other Publications:

1. **Pradyumna Sadhu**, Shantanu Desai (2022).
A test of galaxy cluster fundamental plane for the X-COP sample.
Journal of Cosmology and Astroparticle Physics, 2022. DOI: 10.1088/1475-7516/2022/01/058
2. **Pradyumna Sadhu**, Shantanu Desai (2022).
Characterization of the GRB prompt fundamental plane using Fermi-GBM data.
Journal of High Energy Astrophysics, Vol. 35, pp. 77–82. DOI: 10.1016/j.jheap.2022.06.003
3. **Pradyumna Sadhu**, Shantanu Desai (2021).
A test of Radial Acceleration Relation for the Giles et al Chandra cluster sample.
Physics of the Dark Universe, Vol. 33. DOI: 10.1016/j.dark.2021.100854
4. Jacobson-Galán, W.V. et al. (including **Pradyumna Sadhu**) (2023).
SN 2023ixf in Messier 101: Photo-ionization of Dense, Close-in Circumstellar Material in a Nearby Type II Supernova.
The Astrophysical Journal, Vol. 954. DOI: 10.3847/2041-8213/acf2ec.
Contributed as a co-observer at Lick Observatory when this close and bright supernova went off.

TALKS AND CONFERENCE PRESENTATIONS

- **248th meeting of the American Astronomical Society** June 2026
Pasadena, CA, USA
- **GALFRESCA 2025** Sept 2025
UC San Diego, CA, USA
- **Astronomical Society of India Meeting 2025** Feb 2025
NIT Rourkela, India
- **Seminar on Ultra-faint Dwarf Satellite Population in Galaxy Clusters within Λ CDM** Feb 2025
IIT Hyderabad, India (Invited)
- **Physics and Astronomy Student Seminar** Sept 2024
UC Riverside, CA, USA
- **SoCalDM workshop** May 2024
UC Riverside, CA, USA
- **Innovation in Appalachia: Undergraduate Symposium** Aug 2021
West Virginia University (online due to COVID-19)
- **Presision 2021** Sept 2021
Presidency University, Kolkata, India (online due to COVID-19)

CONFERENCE POSTER PRESENTATIONS

- **IAU Symposium 403: The Hidden Beauty of the Galactic Outskirts** Oct 2025
University of Cordoba, Cordoba, Spain
- **Santa Cruz Galaxies Workshop** Aug 2025
UC Santa Cruz, CA, USA
- **Dwarf Galaxies, Star Clusters, and Streams in the LSST Era Workshop** July 2024
University of Chicago, IL, USA
- **Astronomical Society of India Meeting 2022** Feb 2022
Online due to COVID-19
- **ESO Cluster Mass 2020** Sept 2021
European Southern Observatory (online due to COVID-19)

UNDERGRADUATE MENTORING

- **Sri Devaki Meduri (IIT Hyderabad, India)** May 2026 - present
The Faber-Jackson relation across mass scales in the IllustrisTNG simulation suite.
- **Angela Claibourn (University of Hawaii - Hilo, Hawaii, USA)** June 2025 - Dec 2025
Time evolution of the stellar distributions and kinematics in satellite galaxies

TECHNICAL SKILLS

- **Programming Languages:** Python, C
- **Python Libraries:** pandas, matplotlib, NumPy, SciPy, object-oriented programming, parallel computing
- **Scientific Computing:** HPCC, Bayesian analysis, statistical modeling, data visualization
- **Astronomy Software:** GADGET-2, GADGET-4, AREPO, SMUGGLE
- **Development Tools:** Git, GitHub, ssh, Linux/Unix systems
- **Document Preparation:** LaTeX, scientific writing and typesetting
- **Observational Skills:** GALFIT, Attended Observational Astronomy Workshop at the Lick Observatory (May 2023)

HONORS AND AWARDS

- **Dean's Distinguished Fellowship** 2022
University of California, Riverside
- **Silver Medal** 2022
Indian Institute of Technology, Hyderabad
- **Research Excellence Award** 2022
Indian Institute of Technology, Hyderabad
- **Academic Excellence Award** 2018, 2020
Indian Institute of Technology, Hyderabad

TEACHING AND VOLUNTEER EXPERIENCE

- **Teaching Assistant for PHYS020 Exploring the Universe: an adventure in Astronomy** 2025 - present
University of California, Riverside
 - Holding discussion sections consisting a 30 minute lecture and a 10 minute quiz
 - Creating interest in undergraduate students across different disciplines for astronomy and stressing on its importance
- **Teaching Assistant for Physics Laboratories** 2023 - 2024
University of California, Riverside
 - Teaching PHYS 02LA, 02LB, and 02LC laboratory courses for pre-medical students
 - Guiding students through experimental procedures and data analysis
 - Evaluating student performance and providing constructive feedback
- **Teaching Assistant for Special Relativity and Mathematical Physics** 2021 - 2022
Indian Institute of Technology, Hyderabad
 - Created and graded problem sets; evaluated the finals, for sophomore students
- **Online Math and Physics Tutor** 2022
Akshardaan and National Service Scheme, IIT Hyderabad
 - Taught high school students mathematics and physics online during COVID-19 pandemic
 - Provided educational support to economically limited students during challenging times as a part of the National Service Scheme
- **STEM Summer School Instructor** June 2024
University of California, Riverside
 - Assisted in teaching data analysis using Python to more than 50 high school students
 - Developed hands-on coding exercises

LEADERSHIP AND OUTREACH

- **Co-President, Astronomy Union for Reading, Outreach and Research Activities** 2025 - Present
University of California, Riverside
 - Leading astronomy outreach and research activities for graduate students
 - Organizing seminars, workshops, and public engagement events
 - Fostering collaborative research environment within the astronomy community
- **Public Telescope Observation Volunteer** 2023 - Present
University of California, Riverside
 - Volunteering at multiple public telescope observation sessions
 - Educating community members about celestial objects and astronomical phenomena

- Fostering public interest in astronomy and science
- **Seminar Host for Physics and Astronomy Student Seminars** 2024 - Present
University of California, Riverside
 - Assisting with organization and hosting of regular student seminars
 - Facilitating knowledge sharing among graduate students
 - Creating inclusive and collaborative research environment
- **Founder and Treasurer, Darbar** 2024 - 2026
University of California, Riverside
 - Established student organization for promotion of Indian classical and folk art forms
 - Serving in various leadership positions to organize cultural events
 - Building community and cultural awareness on campus
- **General Secretary, SPIC MACAY IIT-H Subchapter** 2020 - 2022
Indian Institute of Technology, Hyderabad 
 - Led voluntary movement to promote Indian classical art forms among youth
 - Organized cultural events, concerts, and workshops
 - Managed team of volunteers and coordinated with national organization
- **Editor, CosmicVarta** 2023
Indian Astronomy Community Outreach Platform 
 - Editing and curating astronomy research content for public dissemination
 - Promoting Indian astronomy research to broader audiences
 - Bridging gap between research community and general public
- **Judge, Riverside Unified Science and Engineering Fair** Jan 2024
Riverside Unified School District
 - Evaluated student science projects and provided feedback
 - Encouraged young students in scientific inquiry
- **Physics Content Developer** May 2020 - June 2020
Embibe
 - Developed physics educational content for online learning platform
 - Created problem sets for Joint Entrance Exam, one of the most competitive exams in the world